

NE 470/570 Monte Carlo Methods for Radiation Transport

Topic 5c: Variance Reduction for Fixed Source Monte Carlo Particle Transport

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Outline

1. Overview
2. PDF biasing with Russian roulette
3. Particle splitting with Russian roulette
4. Weight-window technique*
5. Integral biasing*
6. Hybrid methodologies*

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1. Overview
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6. Hybrid methodologies*

- The MC method for particle transport works well except for situations in which the probability of the outcome of interest is **low**, then **significant computation time** is needed for achieving a **precise and accurate solution**.
- For example, for a simple 1-D, homogeneous, purely absorbing shield, with a total cross-section of $\Sigma_t = 2 \text{ cm}^{-1}$, and thickness of 10 cm, the probability of transmission is $\sim 2.1 \times 10^{-9}$.
- To achieve a precision of 5%, it is necessary to simulate $\sim 1.9 \times 10^{11}$ particle histories. If we consider that each history requires about 10^{-4} seconds, then this simple simulation requires ~ 220 days of computation time in serial! This demonstrate that need for more clever techniques referred to as **variance reduction (VR)** techniques.

- VR techniques for particle transport are developed based on **changing (biasing) of physical quantities or functions for achieving certain objective with reduced variance.**
- To preserve the expected physical outcome, each particle is given a **statistical weight (w)** that has to be adjusted to compensate for the changed quantity/function.
- This is accomplished by introducing an equation for the preservation of the expected outcome as follows

$$w_{\text{biased}} \times (\text{function/quantity})_{\text{biased}} = w_{\text{unbiased}} \times (\text{function/quantity})_{\text{unbiased}} \quad (1)$$

- The VR techniques used in fixed-source particle transport can be grouped into five categories:
 - 1 PDF biasing
 - 2 particle splitting and rouletting
 - 3 weight-window
 - 4 integral biasing
 - 5 hybrid methodologies
- A good VR technique should result in a **precise and accurate solution in a short time**.
- VR techniques commonly depend on user-defined parameters, which significantly impact the performance and solution accuracy. Hence, any VR technique requires a set of heuristic guidelines and/or theoretical techniques for generation of an appropriate set of parameters.
- The performance of a MC algorithm with VR is measured based on its FOM. A good VR technique is expected to result in significant improvement.
- Note that prior to calculating the FOMs, it is necessary to demonstrate that both algorithms follow **the expected behavior of FOM and relative uncertainty** as discussed in Lecture 5b.

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- To preserve the expected outcome, Equation (1) reduces to

$$w_{\text{biased}} \times \text{PDF}_{\text{biased}} = w_{\text{unbiased}} \times \text{PDF}_{\text{unbiased}} \quad (2)$$

- For example, if the biased PDF is increased, the particle weight is decreased proportionally, so that the expected outcome of the unbiased PDF is preserved.
- There are a few commonly used techniques in this category:
 - 1 Implicit capture or survival biasing with Russian roulette
 - 2 Path length biasing
 - 3 Exponential transformation biasing
 - 4 Forced collision biasing

Outline

2. PDF biasing with Russian roulette

2.1. Implicit capture or survival biasing with Russian roulette

2.2. Path length biasing

2.3. Exponential transformation biasing

2.4. Forced collision biasing

- In a shielding problem, one of the objectives is to **determine the expected detected (survived) particles in the back of the shield**. The number of detected particles, however, can be reduced significantly with the increasing absorption interactions, and consequently significant computation time is needed for achieving a statistically reliable result.
- In the implicit capture technique, **a particle is forced to undergo a scattering interaction, by setting the probability of scattering ($p_s = 1$)**.
- Using Equation (2) for an initial particle weight of w_0 , and the unbiased scattering PDF ($p_s = \frac{\Sigma_s}{\Sigma_t}$), we obtain the following preservation equation:

$$w_1 \times 1 = w_0 \times \frac{\Sigma_s}{\Sigma_t} \quad \rightarrow \quad w_1 = w_0 \frac{\Sigma_s}{\Sigma_t} \quad (3)$$

which indicates that as p_s is set to 1, i.e., p_s is increased by a factor of $\frac{\Sigma_t}{\Sigma_s}$, the particle weight (w) is reduced proportionally by a factor of $\frac{\Sigma_s}{\Sigma_t}$.

- Now, if a particle undergoes n interactions, its weight is reduced as

$$w_n = w_0 \left(\frac{\Sigma_s}{\Sigma_t} \right)^n \quad (4)$$

- Since Σ_s can be much smaller than Σ_t , the above equation indicated that the weight of a particle (w_n) may reduce significantly after several interactions.
- In other words, a low weight particle may have minimal contribution to the particle count and its statistics. In such situations, it is wise to set a **lower weight threshold** w_{cutoff} (referred to as **weight cutoff**), at which the particle fate is determined by applying **the game of Russian roulette**.
- The implicit capture with Russian roulette is an effective VR technique for deep penetration shielding problems.
- The procedure of the Russian roulette technique:
 - 1 After a particle weight has fallen below the weight cutoff w_{cutoff} , a random number is generated and compared to a parameter, e.g., $\frac{1}{d}$, where d is commonly chosen in a range of $[2, 10]$. Then, the fate of the particle is determined according to:
 - 2 If $\eta \leq \frac{1}{d}$, the particle history is **continued** with the particle weight increased by a factor of d , i.e., $w_{\text{new}} = d \times w_{\text{current}}$;
 - 3 Otherwise, the particle history is **terminated**.
- Note that **the Russian Roulette technique can be used with any other VR techniques that encounter very low weight particles**.

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2. PDF biasing with Russian roulette

- 2.1. Implicit capture or survival biasing with Russian roulette
- 2.2. Path length biasing
- 2.3. Exponential transformation biasing
- 2.4. Forced collision biasing

- To increase the chance of particle transport to the region of interest (ROI), one may change the free-flight probability such that the particle path-length is stretched toward the ROI (e.g., back of a shield).
- The FFMC formulation for sampling the path-length (r) indicates that r increases with the decreasing total cross-section.

$$r = -\frac{\ln \eta}{\Sigma_t} \quad (5)$$

- Therefore, in the **path-length biasing** technique, one may consider replacing the total cross-section Σ_t with a smaller quantity such as the scattering cross-section Σ_s .
- This means that the biased PDF for sampling r is expressed by

$$p_{\text{biased}}(r) = \Sigma_s e^{-\Sigma_s r} \quad (6)$$

- Using the biased PDF, the FFMC for sampling path length r reduces to

$$r = -\frac{\ln \eta}{\Sigma_s} \quad (7)$$

- To preserve the expected outcome for the biased case

$$\begin{aligned}
 w_1 \times \Sigma_s e^{-\Sigma_s r} &= w_0 \times \Sigma_t e^{-\Sigma_t r} \\
 w_1 &= w_0 \times \frac{\Sigma_t}{\Sigma_s} e^{-\Sigma_a r}
 \end{aligned} \tag{8}$$

where $\Sigma_a = \Sigma_t - \Sigma_s$.

- Here, indeed, the path length (r) is stretched, but it is done toward *both sides*¹ of the shield. Hence the path length biasing is not very effective, and an alternative method is needed to account for direction of ROI.

¹More precisely, it should be “both directions towards and away from the ROI”.

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- 2.3. Exponential transformation biasing
- 2.4. Forced collision biasing

- To overcome the limitation of path-length biasing, the **exponential transformation** has been developed. This technique provides the ability to **expand the path-length when a particle is moving toward the ROI, while contracting the path-length when the neutron is moving away from the ROI.**
- The biased PDF for the exponential transformation is expressed by²

$$p_{\text{biased}}(r) = (\Sigma_t - c\mu) e^{-(\Sigma_t - c\mu)r} \quad (9)$$

where c is a user-defined positive parameter, and μ is the particle direction cosine.

- In choosing the c parameter, one has to maintain that $\Sigma_t - c\mu > 0$ for all particle directions, i.e., $c < \Sigma_t$ (since the maximum positive value for μ is 1).

²There is a typo in the textbook, Equation (7.11) is $p_{\text{biased}}(r) = (\Sigma_t - c\mu) e^{-(\Sigma_t - \mu)r}$.

- As intended, along the direction of interest ($\mu > 0$), the PDF increases, while it decreases for $\mu < 0$. This is further demonstrated if we derive the corresponding FFMC formulation given by

$$r = -\frac{\ln \eta}{\Sigma_t - c\mu} \quad (10)$$

- Compared to the path-length biasing in Equation (7), Equation (10) yields a larger path-length only in the direction of interest, while it shortens the path-length in the opposite direction.
- To preserve the expected outcome, Equation (2) for exponential transformation reduces to

$$w_1 \times (\Sigma_t - c\mu) e^{-(\Sigma_t - c\mu)r} = w_0 \times \Sigma_t e^{-\Sigma_t r} \quad \rightarrow \quad w_1 = w_0 \times \frac{\Sigma_t}{\Sigma_t - c\mu} e^{-c\mu r} \quad (11)$$

- Now, if a particle goes through n scattering interactions followed by n path-length sampling, the biased weight reduces to

$$w_n = w_0 \cdot e^{-c(x-x_0)} \prod_{i=1}^n \frac{\Sigma_{t,i}}{\Sigma_{t,i} - c\mu_i} \quad (12)$$

where $x - x_0 = \sum_{i=1}^n \mu_i r_i$, and x_0 and x refer to the initial and final positions of the particle on the axis of interest (e.g., x -axis), respectively.

- Note that if $(\Sigma_{t,i} - c\mu_i)$ becomes very small, then the biased weight may become very large, resulting in a large count, and potentially a very biased solution.
- To fix this issue, this method should be combined with a splitting technique such as the weight-window technique that will be discussed later.
- Finally, it is important to note that this technique can be easily combined with the implicit capture technique, as both are effective for a shielding problem.

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- 2.4. Forced collision biasing

- **Forced collision biasing** forces particle collisions in a small volume that otherwise can be missed using the MC particle transport.
- The technique is described in the following steps:

Step 1: A particle of weight (w) enters the volume of interest.

Step 2: The particle is split into two particles (*uncollided* and *collided*) of smaller weights: the *uncollided* particle passes through the volume without any collisions; the *collided* particle is forced to have a collision in the volume.

Step 3: The **weight of the *uncollided* particle** is set to

$$w_{\text{uncollided}} = w e^{-\Sigma_t u} \quad (13)$$

and the **weight of the *collided* particle** is set to

$$w_{\text{collided}} = w (1 - e^{-\Sigma_t u}) \quad (14)$$

Here, u refers to the path-length that spans the volume along the particle direction.

Step 4: The PDF (referred to as $g(r)$) for sampling the path-length of the collided particle within the volume of interest is obtained by setting the preservation equation as follows

$$w_{\text{collided}} \times g(r) = w \times \Sigma_t e^{-\Sigma_t r} \quad (15)$$

Hence, the **PDF for the collided particle** reduces to

$$g(r) = \frac{\Sigma_t e^{-\Sigma_t r}}{1 - e^{-\Sigma_t u}}, \quad \text{where } 0 \leq r \leq u \quad (16)$$

Step 5: Path-length (r) within the volume for the **collided particle** is sampled by using the following FFMC derived based on the PDF $g(r)$.

$$\int_0^r \frac{\Sigma_t e^{-\Sigma_t t}}{1 - e^{-\Sigma_t u}} dt = \frac{1 - e^{-\Sigma_t r}}{1 - e^{-\Sigma_t u}} = \eta \quad (17)$$

The analytical inversion is:

$$r = -\frac{\ln [1 - \eta (1 - e^{-\Sigma_t u})]}{\Sigma_t} \quad (18)$$

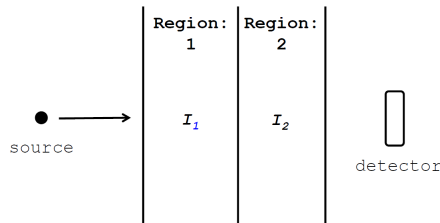
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- The concept of **particle splitting** stems from the fact that **different particles have different levels of contribution (or importance)** to an objective.
- In other words, in a simulation, there are particles of various degrees of *importance*. Therefore, it is desirable to **increase the survivability of the important particles**, and simultaneously judiciously **eliminate less important particles**.
- It is obvious that a splitting technique can be effective only if the importance of particles are selected appropriately. Using erroneous importance may lead to biased results in a rather short time, or waste of computation time by splitting and transporting *unimportant* (or *low weight*) particles.
- In this section, we will discuss particle splitting techniques based on particle attributes including **position (geometry)**, **energy** and **direction (angular)**.

Geometry splitting

- In geometric splitting, the simulation region is partitioned into a number of subregions, and each subregion is given an importance (I) such that **the importance increases as one approaches the region of interest**.
- As a particle moves from a subregion of *low importance* to a subregion of *high importance*, it is split into a number of particles with *lower importance*; thereby, improving the chance of sampling good particles (which are moving toward the ROI).
- Conversely, if a particle travels from a *high importance* subregion to a *low importance* subregion, a game of *Russian roulette* is performed.
- This technique can be explained more clearly via a 1-D shield that is partitioned into two subregions with importances of I_1 and I_2 , respectively.
- If **the objective is to estimate the probability of transmission through the shield**, then it is necessary to set $I_1 < I_2$.



- For a particle of weight w moving in the direction of higher importance, i.e., from subregion 1 to subregion 2, the particle is split into a number of identical particles of lower weights depending on the importance ratio of subregions, i.e., $\frac{I_2}{I_1}$ as follows:

- 1 If $n = \frac{I_2}{I_1}$ is an integer ≥ 2 , the particle is split into n identical particles, each with a weight of $\frac{w}{n}$.
- 2 If $r = \frac{I_2}{I_1}$ is a real number ≥ 2 , a random number η is generated and compared with

$$\Delta = \frac{I_2}{I_1} - \text{INT} \left[\frac{I_2}{I_1} \right] = r - n \quad (19)$$

- (1) If $\eta \leq 1 - \Delta$, the particle is split into n identical particles, each with a weight of $\frac{w}{n}$.
- (2) Otherwise, the particle is split into $(n + 1)$ identical particles, each with a weight of $\frac{w}{n+1}$.

- Actually, scenario 1 is a special case of scenario 2, since when n is an integer, Δ will be 0, $\eta \leq 1 - \Delta$ is always true.

- For example, consider $r = 2.63$, then above procedure indicates that 63% of the time particles are split into 3, and 37% of the time they are split into 2 particles.

- For a particle **moving in the direction of lower importance**, i.e., moving from subregion 2 to subregion 1, a Russian roulette procedure is applied as follows:
 - 1 A random number η is generated;
 - 2 If $\eta \leq \frac{1}{r}$, the particle is kept and its weight is increased to $w \times r$;
 - 3 Otherwise, the particle is killed.
- To assure that the geometric splitting technique is effective, one has to devote significant care in selecting appropriate importance for each subregion.
- A few heuristic recommendations are listed below:
 - 1 Keep the ratio of region importances close to each other (within a factor of 2-4);
 - 2 Consider region sizes of 2 MFP;
 - 3 Maintain the same number of particles for all regions;
 - 4 Do not split particles in a void.

- The major difficulty in using the geometric splitting technique is in assigning the proper importance to subregions. This issue can be significantly remedied, if one knows the **neutron importance (or adjoint) function**.
- Wagner and Haghghat³ have developed methodologies for **automatic generation of the space- and energy-dependent importance distributions**. The importance function will be discussed later.
- It is also worth noting that, in highly angular dependent situations, the effectiveness of geometric splitting may diminish because its importance is not angular-dependent. The use of an **angular-dependent importance function** may remedy this issue.
- Finally, a **zero importance** can be assigned to a region; this is quite useful in representing non-reentrant boundary conditions and/or performing model simplifications/changes.

³ John C Wagner and Alireza Haghghat. Automated variance reduction of MC shielding calculations using the discrete ordinates adjoint function. Nuclear Science and Engineering, 128(2):186-208, 1998.

Energy splitting and Angular splitting

■ Energy splitting

- In some simulations, certain energy ranges are more important than others, thus, we may split particles when their energies lie within the important energy ranges.
- Again Russian roulette is used for particles that may not contribute significantly to the problem objective.
- As an example, if there are two isotopes that compete in neutron capture (at different energies) and one of them is of interest, we split the particles (with “correct” energy) so that we can increase the chance of interaction with the isotope of interest.

■ Angular splitting

- If a particular range of solid angle is important in a simulation, we may split particles when they move into that solid angle.
- Russian roulette is used for particles that move out of the solid angle.

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- The **weight-window technique** combines **space, energy, and angular splitting** techniques with the Russian roulette, but commonly, only space and energy splitting are considered.
- The weight-window provides a unique technique for controlling both high and low particle weights; **high weight particle** may result in biasing the solution, while **low weight particle** may lead to significant inefficiency, as low weight particle may have minimal impact on the results.
- Partitioned into space-energy cells, and a weight window with lower and upper weights, $w_l(\mathbf{r}, E)$ and $w_u(\mathbf{r}, E)$ is assigned to each cell.
- If a particle at position $\mathbf{r}$ with energy E has a weight in the acceptable range of $[w_l, w_u]$, its history is continued; otherwise, it is treated via one of the following procedure:
 - 1 If the particle weight is **less than** w_l , Russian roulette is performed and the particle is either terminated or its weight is increased to a value (w_s) within the acceptable range.
 - 2 If the particle weight is **greater than** w_u , the particle is split into new particles of weight within the acceptable range.

- Generally, w_u and w_s are related to the lower weight w_l . For example, the MCNP code⁴ recommends the following relations:

$$w_u = 5w_l \quad (20)$$

and

$$w_s = 2.5w_l \quad (21)$$

- Commonly, proper selection of an appropriate weight distribution for a complex problem is difficult and requires a significant amount of an analyst's time. The MCNP code offers an iterative procedure for the generation of weights, referred to as the *weight-window generator*^{5,6}.
- However, Haghghat strongly recommends the *use of the importance (adjoint) function methodology for determination of the energy and space-dependent weight distributions*. This is discussed briefly in the following section.

⁴ X-5 MC Team. MCNP-A General Monte Carlo NParticle Transport Code, Version 5, 2003.

⁵ T E Booth. Automatic importance estimation in forward MC calculations. Transactions of the American Nuclear Society, 41, 1982.

⁶ J S Hendricks. A code-generated MC importance function. Transactions of the American Nuclear Society, 41, 1982.

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Importance (adjoint) function methodology

- In a shielding problem, one of the key objectives is determination of the radiation dose or detector response at the surface of a shield. Theoretically, the detector response can be calculated by

$$R = \langle \Sigma_d \psi \rangle \quad (22)$$

where $\langle \cdot \rangle$ is the Dirac sign that refers to **integration over all independent variables** ($\mathbf{r}, E, \hat{\Omega}$), R refers to detector response, Σ_d is detector cross-section, and ψ is the particle angular flux that can be calculated using the linear Boltzmann equation (LBE).

- Another approach to determine the dose or response, is to use the **importance (adjoint) function** methodology. The LBE for a fixed-source problem can be written in an operator form as follows

$$H\psi(p) = q(p), \text{ in } V \quad (23)$$

where p refers to $(\mathbf{r}, E, \hat{\Omega})$, V is the problem volume, ψ is the angular flux, q is the fixed source distribution, and H is the **transport operator** given by⁷

$$H = \hat{\Omega} \cdot \nabla + \Sigma_t - \int_0^\infty dE' \int_{4\pi} d\hat{\Omega}' \Sigma_s(\mathbf{r}, E' \rightarrow E, \hat{\Omega}' \rightarrow \hat{\Omega}) \quad (24)$$

⁷In the textbook, Equation (7.26), $\int_0^\infty dE' \int_{4\pi} d\hat{\Omega}' \Sigma_s(\mathbf{r}, E \rightarrow E', \hat{\Omega} \rightarrow \hat{\Omega}')$ is used, but it should be $\int_0^\infty dE' \int_{4\pi} d\hat{\Omega}' \Sigma_s(\mathbf{r}, E' \rightarrow E, \hat{\Omega}' \rightarrow \hat{\Omega})$.

- Considering a vacuum (non-reentrant) boundary condition, the angular flux for incoming directions is expressed as

$$\psi = 0, \text{ for } \hat{n} \cdot \hat{\Omega} < 0 \text{ and } \mathbf{r} \in \Gamma \quad (25)$$

- The corresponding particle importance equation is given by

$$H^* \psi^*(p) = q^*(p), \text{ in } V \quad (26)$$

where ψ^* is the **particle importance function**, q^* is the **importance source**, and H^* is the **importance operator** given by⁸

$$H^* = -\hat{\Omega} \cdot \nabla + \Sigma_t - \int_0^\infty dE' \int_{4\pi} d\hat{\Omega}' \Sigma_s(\mathbf{r}, E \rightarrow E', \hat{\Omega} \rightarrow \hat{\Omega}') \quad (27)$$

and the importance function for the vacuum boundary condition is expressed as

$$\psi^* = 0, \text{ for } \hat{n} \cdot \hat{\Omega} > 0 \text{ and } \mathbf{r} \in \Gamma \quad (28)$$

⁸In the textbook, Equation (7.29), $\int_0^\infty dE' \int_{4\pi} d\hat{\Omega}' \Sigma_s(\mathbf{r}, E' \rightarrow E, \hat{\Omega}' \rightarrow \hat{\Omega})$ is used, but it should be $\int_0^\infty dE' \int_{4\pi} d\hat{\Omega}' \Sigma_s(\mathbf{r}, E \rightarrow E', \hat{\Omega} \rightarrow \hat{\Omega}')$.

- We can form the **commutation relation** between Equations (23) and (26) to obtain

$$\langle \psi^* H \psi \rangle - \langle \psi H^* \psi^* \rangle = \langle \psi^* q \rangle - \langle \psi q^* \rangle \quad (29)$$

- Considering a vacuum boundary condition, one can show the left-hand side of above equation is equal zero, hence

$$\langle \psi q^* \rangle = \langle \psi^* q \rangle \quad (30)$$

- Now, we consider that the importance source is given by

$$q^* = \Sigma_d \quad (31)$$

- Hence, the detector response (R) equation (Equation (22)) can be written in terms of the importance function as

$$R = \langle \psi^* q \rangle \quad (32)$$

where ψ^* is obtained from Equation (23) for a given detector, and q is a given source distribution.

Source biasing based on the importance sampling

- In a MC shielding simulation, the normalized source distribution (source PDF) is sampled, the sampled particles are transported, then a fraction of survived particles are detected depending on the detector cross-section.
- The detected particles represent the detector response (R) (Equation (32)). This means that if ψ^* was known, then the integral is solved by sampling the normalized source as the PDF. The importance sampling can be very effective to reduce the variance of the sampled integral.
- From importance sampling, the most effective PDF is expressed by

$$\hat{q}(p) = \frac{\psi^*(p)q(p)}{R} \quad (33)$$

- Hence, from this observation we conclude that for a MC shielding simulation, **a more effective PDF is a biased source using an approximate importance function** as follows

$$q_{\text{biased}}(p) = \frac{\psi_a^*(p)q(p)}{R_a} \quad (34)$$

where $\psi_a^*(p)$ is the approximate importance function that is obtained by solving an approximate form of Equation (26), and the corresponding approximate response (R_a) is obtained by solving Equation (32).

- To reserve the expected source, we form the corresponding preservation equation as follows

$$w_{\text{biased}}(p) \times q_{\text{biased}}(p) = w(p) \times q(p) \quad (35)$$

- The **biased source particle weight** is obtained by

$$w_{\text{biased}}(p) = \frac{q(p)}{\frac{\psi_a^*(p)q(p)}{R_a}} = \frac{R_a}{\psi_a^*(p)} \quad (36)$$

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The CADIS methodology

- Over the past 20 years, there has been significant efforts on the development of hybrid methodologies that employ deterministic technique for determination of VR reduction parameters and/or biased functions.
- Here, we will briefly introduce the **CADIS (consistent adjoint driven importance sampling) methodology**^{9,10,11}, and its extension, the **forward-CADIS (FW-CADIS)**^{12,13}.
- The CADIS methodology combines **source biasing** and **space-energy splitting/rouletting** within the **weight-window** technique.
- The biased source and lower weight (w_l) are determined in a consistent manner by using an approximate importance function obtained from a **deterministic discrete-ordinates particle transport** calculation.

⁹ Alireza Haghighat and John C Wagner. Monte Carlo variance reduction with deterministic importance functions. *Progress in Nuclear Energy*, 42(1):25-53, 2003.

¹⁰ John C Wagner. Acceleration of Monte Carlo shielding calculations with an automated variance reduction technique and parallel processing, 1997.

¹¹ John C Wagner and Alireza Haghighat. Automated variance reduction of Monte Carlo shielding calculations using the discrete ordinates adjoint function. *Nuclear Science and Engineering*, 128(2):186-208, 1998.

¹² Scott W Mosher, Aaron M Bevill, Seth R Johnson, Ahmad M Ibrahim, Charles R Daily, Thomas M Evans, John CWagner, Jeffrey O Johnson, and Robert E Grove. ADVANTG - an automated variance reduction parameter generator. ORNL/TM-2013/416, Oak Ridge National Laboratory, 2013.

¹³ John C Wagner, Douglas E Peplow, and Scott W Mosher. FWCADIS method for global and regional variance reduction of Monte Carlo radiation transport calculations. *Nuclear Science and Engineering*, 176(1):37-57, 2014.

- Further, to make the method practical, **only space and energy dependencies** are considered, so we will use **scalar flux ϕ** instead of the **angular flux ψ** . The **biased source** is given by

$$\hat{q}(\mathbf{r}, E) = \frac{\phi_a^*(\mathbf{r}, E)q(\mathbf{r}, E)}{R_a} \quad (37)$$

- And the corresponding **biased weight** for source particles is expressed by

$$\hat{w}(\mathbf{r}, E) = \frac{R}{\phi_a^*(\mathbf{r}, E)} \quad (38)$$

- Using $w_u = 5w_l$ and $w_s = 2.5w_l$, we obtain the lower weight of the weight window as follows

$$w_s(\mathbf{r}, E) = \frac{R_a}{\phi_a^*(\mathbf{r}, E)} \quad (39)$$

then, since w_s is given by

$$w_s = w_l \left(\frac{1+c}{2} \right) \quad (40)$$

so the lower weight is given by

$$w_l(\mathbf{r}, E) = \left(\frac{2}{1+c} \right) \frac{R_a}{\phi_a^*(\mathbf{r}, E)} \quad (41)$$

- Further, during particle transport between $(\mathbf{r}, E)$ and $(\mathbf{r}', E')$ phase spaces, particles are split if the importance ratio $\frac{\phi^*(\mathbf{r}, E)}{\phi^*(\mathbf{r}', E')}$ is greater than 1, otherwise they are rouletted.
- Consequently, the particle weight is adjusted according to

$$w(\mathbf{r}, E) = w(\mathbf{r}', E') \frac{\phi^*(\mathbf{r}', E')}{\phi^*(\mathbf{r}, E)} \quad (42)$$

- It is worth noting that, based on the CADIS methodology, a modified version of MCNP code called A₃MCNP^{14,15} has been developed, which automatically determines the importance function distributions, the biased source, and the weight-window lower weights.
- Wagner has developed a similar algorithm referred to as **ADVANTG**¹⁶.
- Note that the Monaco Monte Carlo code within the SCALE package uses the CADIS formulation through a processing code referred to as MAVRIC¹⁷.

¹⁴ Alireza Haghighat and John C Wagner. Monte Carlo variance reduction with deterministic importance functions. Progress in Nuclear Energy, 42(1):25-53, 2003.

¹⁵ John C Wagner and Alireza Haghighat. Automated variance reduction of Monte Carlo shielding calculations using the discrete ordinates adjoint function. Nuclear Science and Engineering, 128(2):186-208, 1998.

¹⁶ John C Wagner. An automated deterministic variance reduction generator for Monte Carlo shielding applications. In Proceedings of the American Nuclear Society 12th Biennial RPSD Topical Meeting, pages 14-18. Citeseer, 2002.

¹⁷ Douglas E Peplow, Stephen M Bowman, James E Horwedel, and John C Wagner. Monaco/MAVRIC: Computational Resources for Radiation Protection and Shielding in SCALE. TRANSACTIONS-AMERICAN NUCLEAR SOCIETY, 95:669, 2006

The FW-CADIS technique

- CADIS is very effective for the determination of localized objectives, but not so effective for determination of distributed response, e.g., radiation dose distribution throughout a power plant.
- Wagner, Peplow, and Mosher¹⁸ have developed a novel extension to CADIS, referred to as **FW-CADIS**, in which the importance source is weighted inversely by the spatial-dependent forward flux, e.g.,

$$q^*(\mathbf{r}, E) = \frac{\Sigma_d(\mathbf{r}, E)}{\int_0^\infty dE' \Sigma_d(\mathbf{r}, E') \phi(\mathbf{r}, E')} \quad (43)$$

where $\phi(\mathbf{r}, E')$ is the space- and energy-dependent scalar flux.

- This formulation results in **an increase of the adjoint source in regions with low flux values**, and vice versa. In other words, the division by the forward flux increases the importance of regions away from the source.
- FW-CADIS has proved very effective **when the volume is large**, e.g., a nuclear reactor, and it is necessary to determine a certain quantity, e.g., **dose**, throughout the model.
- If necessary, angular dependency can be included. However, more detail can require large amount of memory, and therefore, may prevent the use of the method or reduce the overall benefit.

¹⁸ John C Wagner, Douglas E Peplow, and Scott W Mosher. FWCADIS method for global and regional variance reduction of Monte Carlo radiation transport calculations. Nuclear Science and Engineering, 176(1):37-57, 2014.